

Green Machines + Metallic materials

Year 9 Science · exam sprint · 2 hours

0:00 Lesson 1 · Green Machines

0:50 Lesson 2 · Metallic materials

1:40 Mini mock (both units)

Green Machines (植物) + Metallic materials (金属)

Year 9 科学 · 考前冲刺 · 2 小时

0:00 第一课 · Green Machines (植物)

0:50 第二课 · Metallic materials (金属)

1:40 Mini mock 小模考 (两个单元)

中文版

Dr. Ryan Lei · Auckland 1

本页没有题目。0:00 开始：先发 Classwork；Mini mock 到 1:40 再发。

LESSON 1 · Green Machines

Learning intention: explain how a plant makes food, moves water and makes new plants, using the exam words.

第一课 · Green Machines (植物)

学习目标：用考试用的英文词，解释植物怎样制造食物、运输水分、产生新植物。

中文版

Dr. Ryan Lei · Auckland 2

本页没有题目。这一小时：Loop 1 繁殖 → Loop 2 光合作用 → Loop 3 运输。

Do Now

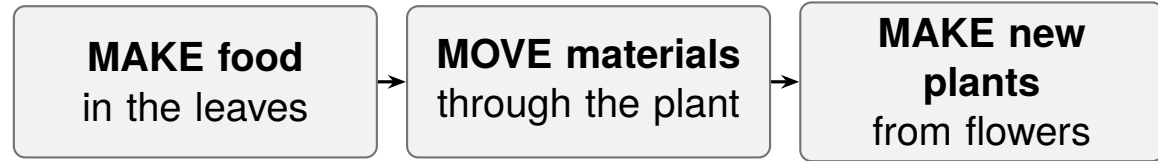
- 1 Which part of a flower makes pollen?
- 2 The pollen nucleus fuses with the ovule nucleus. Is this pollination or fertilisation?
- 3 Name the three conditions a seed needs to germinate.
- 4 Which gas does a leaf take in to make glucose?
- 5 Which tubes carry water up the stem?

Do Now 课前小测

- 1 花的哪一部分产生 pollen (花粉)?
- 2 花粉的 nucleus (细胞核) 和 ovule (胚珠) 的细胞核融合, 这叫 pollination (授粉) 还是 fertilisation (受精)?
- 3 种子 germinate (萌发) 需要哪三个条件?
- 4 叶子吸收哪种气体来制造 glucose (葡萄糖)?
- 5 哪种管道把水沿着茎往上运?

- 1 anther (花药)
- 2 fertilisation (受精)
- 3 water, warmth, oxygen (三个都要说)
- 4 carbon dioxide (二氧化碳)
- 5 xylem (木质部)

The one idea



Every Green Machines question asks about one of these three jobs.
Say which job it is, then use that job's words.

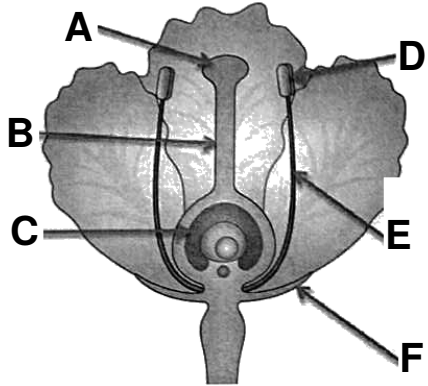
The one idea 一个核心思想



每道 Green Machines 的题，都在问这三件事中的一件。
先判断是哪一件，再用那一件的英文词作答。

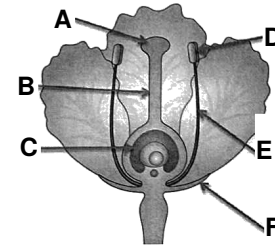
问：用碘液测 starch 属于哪件事？答：MAKE food，因为 starch 是储存的 glucose。

Quick review: Parts of a flower



- 1 Name parts A to F.
- 2 Which two parts make up the stamen?
- 3 Which three parts make up the carpel?
- 4 Which part becomes the fruit?

Quick review 快速回忆：花的结构



- 1 说出 A 到 F 的名称。
- 2 哪两部分组成 stamen (雄蕊)?
- 3 哪三部分组成 carpel (心皮 / 雌蕊)?
- 4 哪一部分以后变成 fruit (果实)?

- 1 A stigma B style C ovary D anther E filament F sepal
2 anther + filament (D 和 E)
3 stigma + style + ovary (A、B、C)
4 ovary (C); ovule 变成 seed

Two ways to make new plants

	Asexual	Sexual
Parents	_____	two
Gametes?	_____	pollen + ovule
Offspring	_____	_____
Examples	runners, bulbs, cuttings	_____

A spider plant grows plantlets on long stems.
Which type of reproduction is this? How do you know?

植物繁殖的两种方式

	Asexual 无性生殖	Sexual 有性生殖
亲本 parents	_____	两个
gametes (配子)?	_____	pollen + ovule
后代 offspring	_____	_____
例子	runners、bulbs、cuttings	_____

(runners 匍匐茎、bulbs 鳞茎、cuttings 扦插)
spider plant (吊兰) 在长茎上长出 plantlets (小植株)。
这是哪种生殖? 你怎么知道?

Asexual: one parent | no gametes | offspring identical (clones)
Sexual: offspring show variation | 例子 flowers and seeds
吊兰: asexual, 因为 one parent、no gametes、plantlets identical to the parent

Why variation matters: write it in this order

- 1 The environment changes, e.g. a new disease arrives.
- 2 Offspring from sexual reproduction are _____ from each other.
- 3 Some offspring have features that let them _____.
- 4 These survive and _____, so the species _____.

A: lines 1–2 **M:** links 2 to 3 **E:** the whole chain + the asexual contrast

Why could a whole field of identical plants die from one disease?

为什么 variation (变异) 重要: 按顺序写

- 1 环境改变, 比如出现一种新疾病。
- 2 有性生殖产生的后代彼此 _____。
- 3 有些后代有能让它们 _____ 的特征。
- 4 这些后代存活并 _____, 所以物种 _____。

A: 写第 1–2 句 **M:** 把 2 和 3 连起来 **E:** 整条链 + 和无性生殖对比

为什么一整片完全一样的植物, 会被一种病全部杀死?

2 different 3 survive (the disease) 4 reproduce / continues
整片死掉: the plants are clones, so if one gets the disease, they all do (没有一株有抵抗力)

From pollen to seed

Put the cards in order.

- A** The pollen tube grows down the style.
- B** Pollen lands on the stigma.
- C** The ovule becomes a seed; the ovary becomes a fruit.
- D** The pollen nucleus fuses with the ovule nucleus.
- E** The pollen tube enters the ovule through the micropyle.
- F** An insect carries pollen away from an anther.

Which cards are pollination? Which card is fertilisation?

从花粉到种子

把卡片按发生的顺序排好。

- A** pollen tube (花粉管) 沿着 style (花柱) 往下长。
- B** 花粉落到 stigma (柱头) 上。
- C** ovule 变成 seed (种子); ovary (子房) 变成果实。
- D** 花粉的细胞核和胚珠的细胞核融合。
- E** 花粉管通过 micropyle (珠孔) 进入胚珠。
- F** 昆虫从 anther (花药) 带走花粉。

哪几张卡片是 pollination? 哪一张是 fertilisation?

顺序: F → B → A → E → D → C

pollination = F 和 B

fertilisation = D

Wind or insect pollination?

Part	Wind-pollinated	Insect-pollinated
Petals	small, dull, no scent	
Stigma		sticky, inside the flower
Anthers	hang outside the flower	
Pollen		larger, sticky or spiky

Full marks: the feature **and** how it helps.

Why do the anthers of grass hang outside the flower?

风媒授粉还是虫媒授粉？

部位	Wind 风媒花	Insect 虫媒花
petals 花瓣	小、不鲜艳、没香味	
stigma 柱头		有黏性、在花里面
anthers 花药	挂在花的外面	
pollen 花粉		较大、黏或带刺

满分要写：特征 **和** 这个特征怎样帮助授粉。
为什么草的 anthers 挂在花的外面？

Insect petals: large, brightly coloured, scented, with nectar
Wind stigma: large and feathery, hanging outside
Insect anthers: firmly attached inside the flower
Wind pollen: small, light, made in huge amounts
草的 anthers: so the wind can blow the pollen away

Spot the error (1)

A strawberry plant makes new plants from runners and from its flowers. Compare the two.

Sam's answer

Runners: the new plant gets gametes from the parent, so it is a bit different from it. Flowers: pollen is carried from the stigma to the anther. This is called fertilisation. The seeds show variation because two parents are involved.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error (1) 找错误

草莓用 runners (匍匐茎) 和花两种方式产生新植株。比较这两种方式。

Sam's answer

Runners: the new plant gets gametes from the parent, so it is a bit different from it. Flowers: pollen is carried from the stigma to the anther. This is called fertilisation. The seeds show variation because two parents are involved.

Sam 的答案 (英文) 有 3 个错误。找出每一个, 改正, 并说明为什么会丢分。

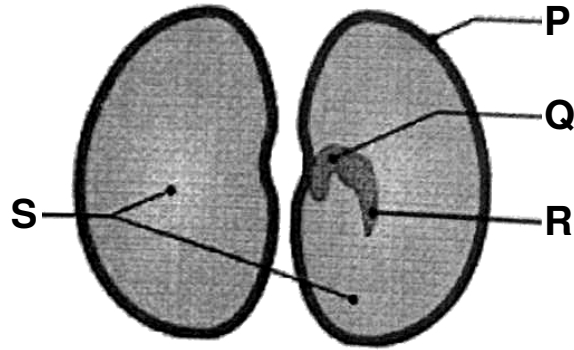
1 runners have no gametes: the new plant is identical (a clone)

2 pollen goes from the anther to the stigma (Sam 写反了)

3 that is pollination, not fertilisation

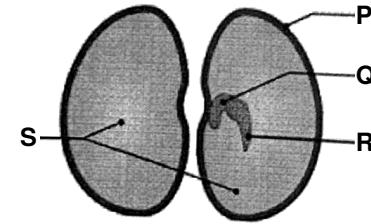
最后一句 (两个亲本 → variation) 是对的

Quick review: Seeds



- 1 Name parts P to S.
- 2 Name the three conditions for germination.
- 3 Why are seeds dispersed far from the parent plant?

Quick review 快速回忆：种子



- 1 说出 P 到 S 的名称。
- 2 说出 germination (萌发) 的三个条件。
- 3 为什么种子要被 dispersed (传播) 到离母株很远的地方？

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1 P testa Q plumule R radicle S cotyledon

2 water, warmth, oxygen

3 so it does not compete with the parent for light, water, minerals and space

Practice

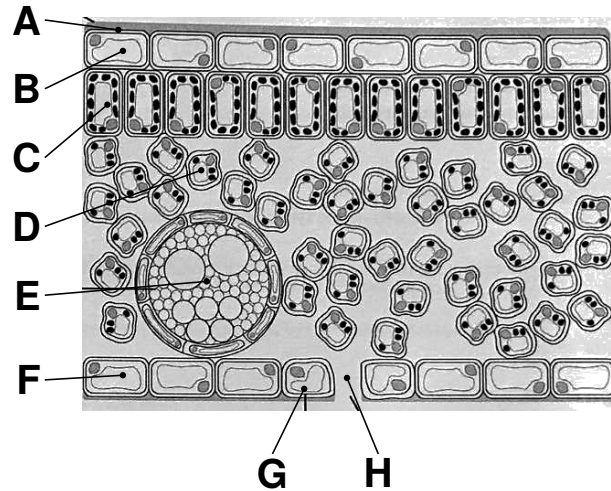
- Q1** A lily has large white petals, a strong scent and nectar. Is it wind- or insect-pollinated? Give two reasons. **[A]**
- Q2** A grower takes cuttings from her best rose bush instead of planting its seeds. Explain why. **[M]**
- Q3** Cress seeds are put in four tubes at 20 °C: A on dry cotton wool; B on wet cotton wool; C on wet cotton wool in a fridge; D under boiled water with a layer of oil on top. Which tubes germinate? Explain the result for D. **[E]**

Practice 练习

- Q1** 百合花有大的白色花瓣、浓香味和 nectar (花蜜)。它是风媒还是虫媒授粉？给两个理由。 **[A]**
- Q2** 种花的人从最好的玫瑰上剪枝 cuttings (扦插), 而不是种它的种子。为什么？ **[M]**
- Q3** 水芹种子放进四支试管 (20 °C): A 干棉花; B 湿棉花; C 湿棉花放冰箱; D 泡在煮过的水里, 上面盖一层油。哪几管会萌发？解释 D 的结果。 **[E]**

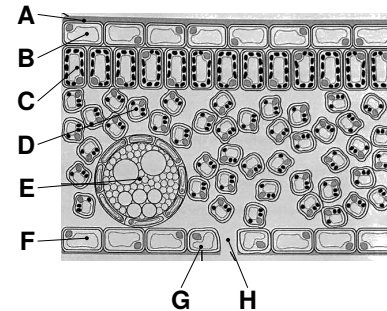
- Q1 insect: large white petals and scent attract insects; nectar is food for them**
Q2 cuttings are asexual → identical plants keep the best features; seeds show variation
Q3 only B. D has no oxygen, so the seed cannot respire to release energy for growth
(A 没有水; C 太冷)

Quick review: Leaf structure



- 1 Name A to H.
- 2 Which layer does most of the photosynthesis?

Quick review 快速回忆：叶片结构



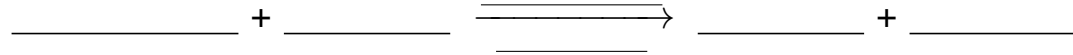
- 1 说出 A 到 H 的名称。
- 2 哪一层进行最多的 photosynthesis (光合作用)?

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- 1 A waxy cuticle B upper epidermis C palisade layer D spongy layer
E vein (xylem + phloem) F lower epidermis G guard cell H stoma
2 palisade layer (C): most chloroplasts

Photosynthesis: the word equation



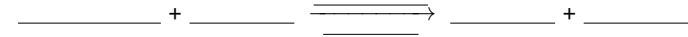
Where does each starting substance enter the plant?

carbon dioxide: through the in the leaf

water: root hairs → → leaf

What does chlorophyll do in this reaction?

Photosynthesis 光合作用：文字方程式



每种原料从哪里进入植物？

carbon dioxide (二氧化碳): 通过叶子上的

water (水): root hairs (根毛) → → 叶子

chlorophyll (叶绿素) 在这个反应里起什么作用？

carbon dioxide + water → glucose + oxygen

箭头上: light energy 箭头下: chlorophyll

CO₂ 通过 stomata 进入; water: root hairs → xylem → leaf

chlorophyll absorbs light energy

Respiration: the same substances, reversed

glucose + _____ → _____ + _____ (+ energy)

	Photosynthesis	Respiration
Day	_____	_____
Night	_____	_____

Which process happens in every living cell, all the time?

Respiration 呼吸作用：同样的物质，反过来

glucose + _____ → _____ + _____ (+ energy 能量)

	Photosynthesis 光合作用	Respiration 呼吸作用
白天	_____	_____
晚上	_____	_____

哪个过程在每一个活细胞里、任何时候都在进行？

glucose + oxygen → carbon dioxide + water (+ energy)
白天：photosynthesis 进行，respiration 进行
晚上：photosynthesis 不进行，respiration 进行
答：respiration (每个活细胞，一直都在进行)

Testing a leaf for starch

Give the reason for each step.

- 1 Put the leaf in boiling water for 1 minute.
- 2 Put it in hot ethanol, standing in a beaker of hot water.
- 3 Rinse it in water.
- 4 Add iodine solution. Blue-black means _____.

Why is the ethanol heated in hot water, not over a Bunsen flame?

测试叶子里有没有 starch (淀粉)

说出每一步的原因。

- 1 把叶子放进沸水 1 分钟。
- 2 放进热的 ethanol (乙醇), 乙醇管放在一杯热水里。
- 3 用水冲洗。
- 4 滴 iodine (碘液)。变成蓝黑色说明 _____。

为什么乙醇用热水加热, 而不是放在 Bunsen (本生灯) 火焰上?

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1 kills the leaf (stops its reactions)

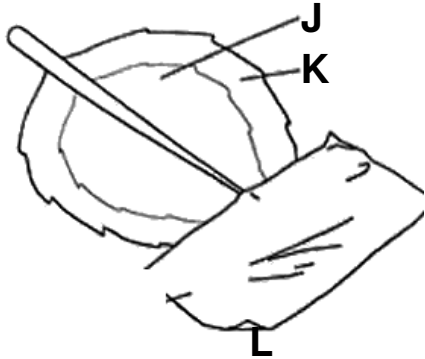
2 removes the chlorophyll, so the colour change can be seen

3 softens the leaf / washes off the ethanol

4 starch is present

ethanol is flammable (易燃), 不能靠近火焰

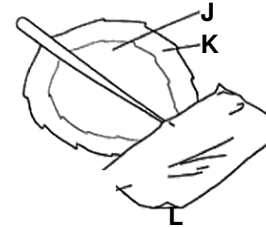
Workbook: the foil leaf



A variegated leaf: J green, K white.
Foil L covered part of it for 2 days in the light. Then the leaf was tested with iodine.

Which areas turn blue-black? Explain each area.

Workbook 练习册题：盖锡箔的叶子



一片 variegated leaf (斑叶, 有绿有白): J 绿色, K 白色。锡箔 L 盖住一部分, 在光下放 2 天, 然后用碘液测试。

哪些部分变蓝黑? 逐个解释。

J (绿色、没盖): **blue-black —chlorophyll + light → photosynthesis → starch**
K (白色): **stays brown —no chlorophyll**
L (盖锡箔): **stays brown —no light**

Spot the error (2)

Write the word equation for photosynthesis. Then say how you would show that a leaf has made food.

Sam's answer

oxygen + water makes glucose + carbon dioxide (light above the arrow). Boil the leaf, put it in hot ethanol, rinse it, then add iodine. If it goes blue-black, the leaf has made glucose. Plants photosynthesise in the day and respire at night.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error (2) 找错误

写出光合作用的文字方程式，再说明怎样证明叶子制造了食物。

Sam's answer

oxygen + water makes glucose + carbon dioxide (light above the arrow). Boil the leaf, put it in hot ethanol, rinse it, then add iodine. If it goes blue-black, the leaf has made glucose. Plants photosynthesise in the day and respire at night.

Sam 的答案（英文）有 3 个错误。找出每一个，改正，并说明为什么会丢分。

1 carbon dioxide + water → glucose + oxygen (O₂ 和 CO₂ 写反了)

2 blue-black shows starch, not glucose

3 plants respire all the time, day and night

实验步骤那一句是对的

Practice

Q1 Two potted plants are kept in the light for 2 days, each under a sealed bell jar. Jar 1 also holds a dish of soda lime, which absorbs carbon dioxide. Jar 2 holds a dish of water. A leaf from each plant is tested with iodine. Predict and explain the results. **[M]**

Q2 In a greenhouse full of plants, the carbon dioxide level rises at night and falls during the day. Explain why. **[E]**

Practice 练习

Q1 两盆植物在光下放 2 天，各罩在一个密封的 bell jar (钟罩) 里。1 号罩里还放了一碟 soda lime (碱石灰，会吸收二氧化碳)；2 号罩里放一碟水。各取一片叶子做碘液测试。预测并解释结果。 **[M]**

Q2 种满植物的温室里，二氧化碳浓度晚上升高、白天下降。为什么？ **[E]**

**Q1 plant 1 stays brown: no carbon dioxide → no photosynthesis → no starch
plant 2 turns blue-black; jar 2 is the control**

**Q2 night: only respiration, gives out CO₂
day: photosynthesis takes in CO₂ faster than respiration gives it out**

Quick review: Transport words

Match each word to its job.

- | | |
|----------------------|---|
| 1 xylem | A carries glucose from the leaves |
| 2 phloem | B lets gases and water vapour in and out of the leaf |
| 3 root hair | C absorbs water from the soil |
| 4 stomata | D carries water and minerals up the plant |
| 5 guard cells | E change shape to open and close the stomata |

Quick review 快速回忆：运输相关的词

把每个词和它的作用配对。

- | | |
|-----------------------------|---------------------------------|
| 1 xylem (木质部) | A 把 glucose 从叶子运走 |
| 2 phloem (韧皮部) | B 让气体和水蒸气进出叶子 |
| 3 root hair (根毛) | C 从土壤里吸水 |
| 4 stomata (气孔) | D 把水和 minerals (矿物质) 往上运 |
| 5 guard cells (保卫细胞) | E 改变形状，打开和关闭气孔 |

1 D 2 A 3 C 4 B 5 E

How water travels through a plant

soil water

↓ by _____

root hair cell

↓

_____ in the root and up the stem

↓

leaf cells: water evaporates

↓

out through the _____ as water vapour

What is the loss of water vapour from the leaves called?

水在植物里怎么走

土壤里的水

↓ 通过 _____

根毛细胞

↓

根里和茎里的 _____

↓

叶细胞：水蒸发

↓

以水蒸气的形式从 _____ 出去

叶子失去水蒸气的过程叫什么？

osmosis → xylem → stomata
叶子失去水蒸气：transpiration (蒸腾作用)

Why the water moves up: write it in this order

A celery stalk stands in blue dye for 2 days. The leaves turn blue.

- 1 Water _____ from the leaves through the stomata.
- 2 This _____ more water up the _____.
- 3 The dye travels with the water, so the _____ turn blue.
- 4 Cut the stalk: the blue dots show where the _____ are.

A: dye goes up **M:** names xylem **E:** transpiration causes the pull

水为什么往上走：按顺序写

芹菜放在蓝色色素水里 2 天，叶子变蓝。

- 1 水从叶子通过 stomata _____。
- 2 这会把更多的水沿着 _____ 往上 _____。
- 3 色素跟着水走，所以 _____ 变蓝。
- 4 把茎切开：蓝点显示 _____ 的位置。

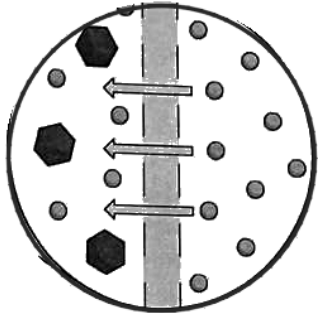
A: 色素往上走 **M:** 说出 xylem **E:** transpiration (蒸腾作用) 造成拉力

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1 evaporates 2 pulls / xylem 3 leaves 4 xylem (vessels)
不能写 “the plant sucks / is thirsty”

Osmosis

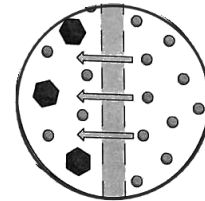


Left: sugar solution. Right: water.

- 1 Which shape is a water molecule?
- 2 Which way do the water molecules move?
- 3 Why can the sugar molecules not cross?

Osmosis is the movement of _____ from a _____ concentration of water to a _____ concentration of water through a _____ membrane.

Osmosis 渗透作用



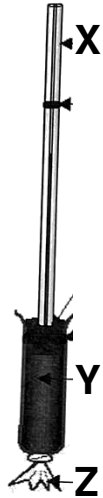
左边：糖溶液。右边：水。

- 1 哪种形状是水分子？
- 2 水分子往哪边移动？
- 3 为什么糖分子过不去？

Osmosis 是 _____ 从水浓度 _____ 的地方到水浓度 _____ 的地方，穿过 _____ 膜的移动。

- 1 the small circles (六边形是 sugar)
 - 2 right to left, into the sugar solution
 - 3 the sugar molecules are too large to fit through the membrane
- 定义空格: water / high / low / semi-permeable

Workbook: the Visking tubing model



The tubing Z sits in a beaker of water. Over 30 minutes the liquid rises up X.

- 1 Which part acts like the root hair? Which acts like the stem?
- 2 Explain why the liquid rises.
- 3 A root hair is long and thin. How does this help it?

Workbook 练习册题：透析袋模型



Visking tubing (透析袋) Z 放在一杯水里，30 分钟内液体沿着 X 管上升。

- 1 哪一部分相当于根毛？哪一部分相当于茎？
- 2 解释液体为什么上升。
- 3 根毛又长又细，这有什么好处？

- 1 Z (Visking tubing) = root hair; X (capillary tube) = stem**
2 water moves into the sucrose solution by osmosis through the membrane, so the liquid is pushed up
3 large surface area, so it absorbs more water

Spot the error (3)

A white carnation stands in blue dye. A day later its petals are blue. Explain.

Sam's answer

The blue water was carried up the phloem tubes. The flower sucks the water up because it is thirsty. Water always gets into a plant by osmosis, from a low concentration of water to a high concentration of water.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error (3) 找错误

一朵白色康乃馨插在蓝色色素水里，一天后花瓣变蓝。解释原因。

Sam's answer

The blue water was carried up the phloem tubes. The flower sucks the water up because it is thirsty. Water always gets into a plant by osmosis, from a low concentration of water to a high concentration of water.

Sam 的答案（英文）有 3 个错误。找出每一个，改正，并说明为什么会丢分。

1 the water goes up the xylem, not the phloem

2 not “sucks / thirsty” : water evaporates through the stomata (transpiration) and pulls water up

3 osmosis is from a high to a low concentration of water (Sam 写反了)

Practice

Q1 Name the process that moves water into a root hair, and the tubes that carry it to the leaves. **[A]**

Q2 In a Visking tubing model the liquid is 12 mm up the tube at the start, 30 mm after 10 minutes and 48 mm after 20 minutes.

(a) Calculate the average rate of rise in mm per minute.

(b) Explain why the liquid rises. **[M]**

Q3 A plant wilts when its roots are put in very salty water. Explain why. **[E]**

Practice 练习

Q1 说出把水送进根毛的过程，以及把水运到叶子的管道。 **[A]**

Q2 透析袋模型里，液体一开始在管里 12 mm 高，10 分钟后 30 mm，20 分钟后 48 mm。

(a) 计算平均上升速度（每分钟多少 mm）。

(b) 解释液体为什么上升。 **[M]**

Q3 把植物的根放进很咸的水里，植物会 wilt（萎蔫）。为什么？ **[E]**

Q1 osmosis; xylem

Q2 (a) $(48 - 12) \div 20 = 1.8$ mm per minute (不是 $48 \div 20$)

Q2 (b) water enters the sugar solution by osmosis, so the liquid rises

Q3 salty water has a lower concentration of water → water moves out of the roots by osmosis → wilts

Ways to lose marks: Green Machines

- L1** Pollination moves pollen; fertilisation fuses the nuclei.
- L2** Asexual: one parent, no gametes, identical offspring.
- L3** Feature **and** how it helps.
- L4** Light and chlorophyll go above and below the arrow.
- L6** Iodine shows starch, not glucose.
- L7** Xylem carries water; phloem carries glucose.

最常见的丢分点: Green Machines

- L1** pollination 送花粉; fertilisation 是两个细胞核融合。
- L2** asexual: 一个亲本、没有配子、后代一样。
- L3** 写特征, **也要**写它怎样帮助。
- L4** light 和 chlorophyll 写在箭头的上下。
- L6** 碘液测的是 starch, 不是 glucose。
- L7** xylem 运水; phloem 运 glucose。

Green Machines p.2

Dr. Ryan Lei · Auckland 27

让她每条举一个今天出现过的例子: **L1 Sam (1); L2 吊兰; L4、L6 Sam (2); L7 康乃馨**

Summary: Green Machines

Flowers: pollination moves pollen; fertilisation joins the nuclei.

Seeds need water, warmth and oxygen; dispersal cuts competition.

Leaves: carbon dioxide + water make glucose + oxygen, using light.

Water: osmosis into root hairs, up the xylem, out by transpiration.

Summary 总结: Green Machines

花: pollination 送花粉; fertilisation 合细胞核。

种子需要水、温暖、氧气; 传播减少竞争。

叶子: carbon dioxide + water 用光变成 glucose + oxygen。

水: osmosis 进根毛, 沿 xylem 上去, transpiration 出去。

本页没有题目。让她闭眼说一遍这四句，再看屏幕对照。

Classwork: Green Machines

On your classwork sheet, about 10 minutes:

- A** Flowers and seeds
- B** Photosynthesis and respiration
- C** Transport and osmosis, with a graph to draw

Write in full sentences for every **explain** question.

Classwork 课堂练习: Green Machines

在课堂练习纸上做，大约 10 分钟：

- A** 花和种子
- B** 光合作用和呼吸作用
- C** 运输和渗透，有一张图要画

每道 **explain** 题都写完整的英文句子。

做 **Classwork Part A–C**，约 10 分钟；答案在 **classwork_answers.pdf**
Part C 的图：0–25 分钟直线上升，之后变平

LESSON 2 · Metallic materials

Learning intention: link properties to uses, calculate density, and explain alloys, rusting and the reactions of metals.

第二课 · Metallic materials (金属)

学习目标：把金属的性质和用途联系起来，计算 density (密度)，解释 alloys (合金)、rusting (生锈) 和金属的化学反应。

本页没有题目。后半本 booklet 她没写：每页先让她说，再补充。

Do Now

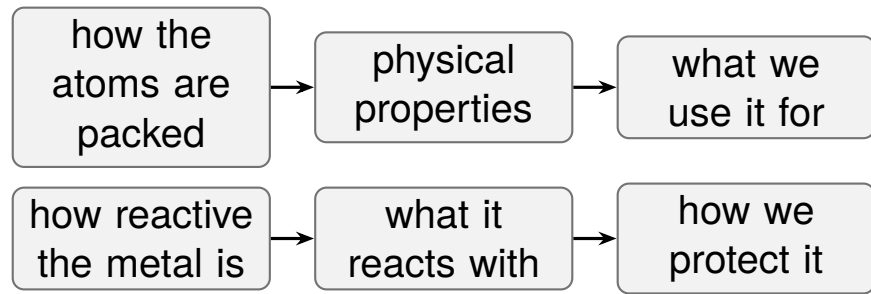
- 1 Give two physical properties shared by most metals.
- 2 Name the three common magnetic metals.
- 3 Write the formula for density.
- 4 Gallium melts at 30 °C. Is it solid or liquid on a 35 °C day?
- 5 A metal is heated red hot, then cooled slowly. Name this heat treatment.

Do Now 课前小测

- 1 说出大多数金属共有的两个物理性质。
- 2 说出三种常见的有磁性的金属。
- 3 写出密度的公式。
- 4 gallium (镓) 在 30 °C 熔化。35 °C 的天气里它是固体还是液体？
- 5 把金属烧到红热，再慢慢冷却。这种热处理叫什么？

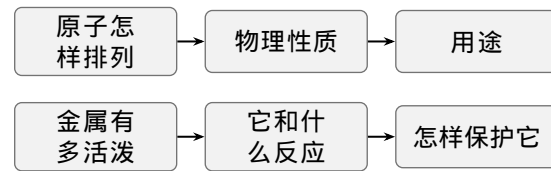
- 1 任选两个：lustrous, malleable, ductile, sonorous, good conductor, high melting point
- 2 iron, nickel, cobalt
- 3 density = mass ÷ volume
- 4 liquid (35 °C 高于熔点)
- 5 annealing (慢慢冷却); quenching 是快速冷却

The one idea



Top row: physical properties. Bottom row: chemical reactions.
Every Metals question sits on one of these two rows.

The one idea 一个核心思想



上排：物理性质。下排：化学反应。
每道金属题都在这两排中的一排。

问：为什么金子做首饰？答：lustrous（上排）+ very unreactive, does not corrode（下排）

Quick review: Property words

Match each word to its meaning, then spell it.

- | | |
|--------------------|---|
| 1 malleable | A has a shiny surface that reflects light |
| 2 ductile | B can be hammered into shape |
| 3 lustrous | C makes a ringing sound when hit |
| 4 sonorous | D can be stretched into wires |
| 5 conductor | E lets heat or electric current pass through |

Quick review 快速回忆：性质词

把每个词和意思配对，然后拼写出来。

- | | |
|--------------------|--------------------|
| 1 malleable | A 表面有光泽，能反光 |
| 2 ductile | B 能锤打成形 |
| 3 lustrous | C 敲击时会发出响声 |
| 4 sonorous | D 能拉成细丝 |
| 5 conductor | E 让热或电流通过 |

Metals p.2, p.20

Dr. Ryan Lei · Auckland 33

1 B 2 D 3 A 4 C 5 E
拼写：malleable (两个 l, -eable)

Choosing a metal

Metal	Melting point (°C)	Strength	Electrical conductivity	Density (g/cm ³)
W	1450	high	low	7.8
X	650	low	high	2.7
Y	1080	medium	very high	8.9
Z	230	low	medium	7.3

Which metal for (a) a bridge frame, (b) wiring in a house, (c) a light overhead power cable? (d) Which would melt in a Bunsen flame (about 800 °C)?

选择金属

金属	熔点 (°C)	强度	导电性	密度 (g/cm ³)
W	1450	高	低	7.8
X	650	低	高	2.7
Y	1080	中	很高	8.9
Z	230	低	中	7.3

哪种金属适合 (a) 桥的框架, (b) 家里的电线, (c) 要求轻的高架电缆? (d) 哪几种会在本生灯火焰 (约 800 °C) 里熔化?

- (a) W: high strength, high melting point
- (b) Y: very high conductivity
- (c) X: high conductivity and low density (2.7) → light
- (d) X 和 Z (熔点低于 800 °C)

Justify a choice: write it in this order

- 1 Name the metal.
- 2 Give the property that matters for this use.
- 3 Say why that property matters for the use.
- 4 Compare: why not another metal? Use the data.

A: 1 + 2 **M:** 1 to 3 **E:** 1 to 4 with numbers

For the power cable I choose _____ because it _____.

This matters because _____.

Y conducts better, but _____.

说明选择的理由：按顺序写

- 1 说出金属的名字。
- 2 说出这个用途需要的性质。
- 3 说明这个性质为什么对这个用途重要。
- 4 比较：为什么不用另一种金属？用数据。

A: 1 + 2 **M:** 1 到 3 **E:** 1 到 4，带数字

高架电缆我选 _____，因为它 _____。

这很重要，因为 _____。

Y 导电更好，但是 _____。

**I choose X because it has a high electrical conductivity and a low density of 2.7 g/cm^3 .
This matters because a light cable puts less weight on the pylons and does not sag.
Y conducts better, but its density is 8.9 g/cm^3 , more than three times heavier.**

Density: the formula and the volume

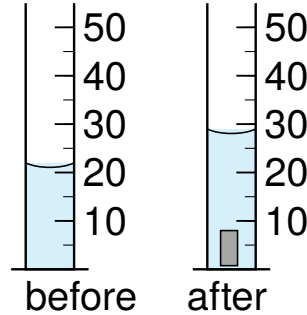
density = $\frac{\text{mass}}{\text{volume}}$

units: g/cm^3 or g/mL

Volume of a regular block:

Volume of an odd-shaped object:
the _____ method

What is the volume of the object on
the right?



Density 密度：公式和体积

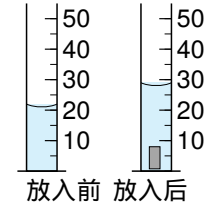
density = $\frac{\text{mass}}{\text{volume}}$

单位: g/cm^3 或 g/mL

规则形状物体的体积:

不规则形状物体的体积:
用 _____ 法

右边物体的体积是多少?



density = mass ÷ volume; 单位 g/cm^3 或 g/mL

规则物体: length × width × height

不规则物体: displacement (排水法)

右边物体: 29 - 22 = 7 mL

Density: worked example

A brass key has a mass of 25.5 g. When it is lowered into a measuring cylinder, the water level rises from 30.0 mL to 33.0 mL.

Step 1 volume = _____ - _____ = _____ mL

Step 2 density = _____ ÷ _____

Step 3 density = _____ with its unit

Check: is your answer bigger than 1 g/mL? Brass sinks, so it should be.

Density 密度：例题

一把 brass (黄铜) 钥匙质量 25.5 g。放进量筒后，水面从 30.0 mL 升到 33.0 mL。

第 1 步 体积 = _____ - _____ = _____ mL

第 2 步 密度 = _____ ÷ _____

第 3 步 密度 = _____ (写上单位)

检查：答案比 1 g/mL 大吗？黄铜会下沉，所以应该更大。

第 1 步：33.0 - 30.0 = 3.0 mL

第 2 步：25.5 ÷ 3.0

第 3 步：8.5 g/mL

Spot the error (4)

A metal bolt has a mass of 39.5 g. When it is put into a measuring cylinder, the water rises from 20.0 mL to 25.0 mL. Calculate the density of the bolt.

Sam's answer

volume = 25.0 mL
density = $39.5 \times 25.0 = 987.5 \text{ g}$

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error (4) 找错误

一个金属螺栓质量 39.5 g。放进量筒后，水面从 20.0 mL 升到 25.0 mL。计算螺栓的密度。

Sam's answer

volume = 25.0 mL
density = $39.5 \times 25.0 = 987.5 \text{ g}$

Sam 的答案有 3 个错误。找出每一个，改正，并说明为什么会丢分。

1 volume = $25.0 - 20.0 = 5.0 \text{ mL}$ ，不是 25.0

2 density = mass $\div$ volume，不是乘

3 unit is g/mL，不是 g

正确: $39.5 \div 5.0 = 7.9 \text{ g/mL}$

Practice

Q1 A metal block is $2.0\text{ cm} \times 2.0\text{ cm} \times 3.0\text{ cm}$ and has a mass of 64.8 g . Calculate its density. **[A]**

Q2 A ring has a mass of 31.5 g . It makes the water in a measuring cylinder rise from 40.0 mL to 43.0 mL . Which metal is it? **[M]**
iron 7.9 copper 8.9 silver 10.5 aluminium 2.7 (g/mL)

Q3 Silver conducts electricity better than copper. Explain why house wiring is made of copper. **[E]**

Practice 练习

Q1 一块金属 $2.0\text{ cm} \times 2.0\text{ cm} \times 3.0\text{ cm}$ ，质量 64.8 g 。计算它的密度。 **[A]**

Q2 一枚戒指质量 31.5 g ，放进量筒后水面从 40.0 mL 升到 43.0 mL 。它是哪种金属？ **[M]**
铁 iron 7.9 铜 copper 8.9 银 silver 10.5 铝 aluminium 2.7 (g/mL)

Q3 银的导电性比铜好。解释为什么家里的电线用铜做。 **[E]**

Metals p.4, p.6–7

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Q1 $2.0 \times 2.0 \times 3.0 = 12.0\text{ cm}^3$; $64.8 \div 12.0 = 5.4\text{ g/cm}^3$

Q2 $43.0 - 40.0 = 3.0\text{ mL}$; $31.5 \div 3.0 = 10.5\text{ g/mL} \rightarrow \text{silver}$

Q3 copper is almost as good a conductor, much cheaper, and ductile (can be drawn into wires)

Quick review: Heat treatments

- 1 Heat red hot, then cool quickly in cold water.
- 2 Heat red hot, then cool slowly.
- 3 Heat strongly until a blue sheen appears.

Name each treatment: annealing, quenching or tempering.

Then match it to a result: **A** soft, easy to bend **B** hard but brittle
C tough and springy

- 4 A hair clip must spring back into shape. Which treatment does it need?

Quick review 快速回忆：热处理

- 1 烧到红热，然后放进冷水里快速冷却。
- 2 烧到红热，然后慢慢冷却。
- 3 强烈加热，直到出现蓝色光泽。

说出名称：annealing (退火)、quenching (淬火) 或 tempering (回火)。

再配上结果：**A** 软、容易弯 **B** 硬但脆 **C** 有韧性、有弹性

- 4 发夹要能弹回原来的形状，需要哪种热处理？

Metals p.8

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1 quenching → B (硬但脆)

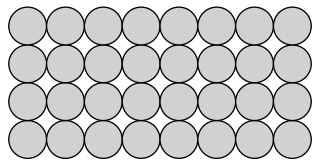
2 annealing → A (软)

3 tempering → C (有韧性、有弹性)

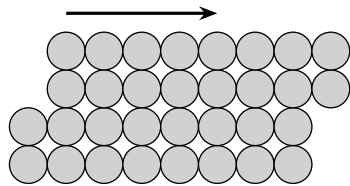
4 tempering (发夹要 springy)

Why alloys are harder (1)

A pure metal: every atom is the same size, in regular layers.



before

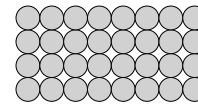


after a push

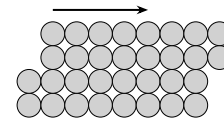
What did the layers do when they were pushed? So is a pure metal hard or soft?

合金为什么更硬 (1)

纯金属：每个原子都一样大，排成整齐的层。



推之前



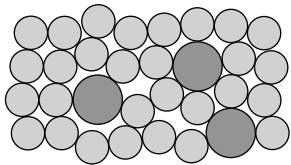
推一下以后

推的时候，这些层怎么了？所以纯金属是硬还是软？

**the layers slide over each other easily
so a pure metal is soft (所以 malleable、ductile)**

Why alloys are harder (2)

An alloy: a metal mixed with another element.

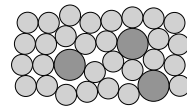


- 1 The atoms are _____ sizes.
- 2 This _____ the regular layers.
- 3 The layers cannot _____ easily.
- 4 So the alloy is _____ than the pure metal.

A: harder **M:** + different-sized atoms **E:** whole chain

合金为什么更硬 (2)

alloy (合金): 一种金属和另一种元素混合。



- 1 原子的大小 _____。
- 2 这 _____ 了整齐的层。
- 3 层不能轻易地 _____。
- 4 所以合金比纯金属 _____。

A: 更硬 **M:** + 原子大小不同 **E:** 整条链

1 different 2 disrupts 3 slide over each other 4 harder

Alloys in use

Alloy	Main metals	Use
bronze	copper + _____	gears, machine bearings
solder	tin + lead	_____
stainless steel	iron, chromium, nickel	_____
nitinol	nickel + titanium	_____

Why is solder good for joining electrical wires?
Is an alloy a compound or a mixture?

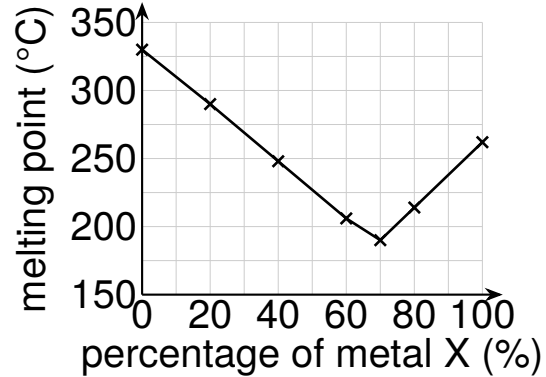
合金的用途

合金	主要金属	用途
bronze 青铜	铜 + _____	齿轮、机器轴承
solder 焊锡	锡 + 铅	_____
stainless steel 不锈钢	铁、铬、镍	_____
nitinol 镍钛合金	镍 + 钛	_____

为什么焊锡适合连接电线？
合金是 compound (化合物) 还是 mixture (混合物)？

bronze: tin solder: joining electrical wires
stainless steel: cutlery, sinks (不生锈) nitinol: braces, glasses frames (shape memory)
solder: low melting point alloy 是 mixture (混合物)

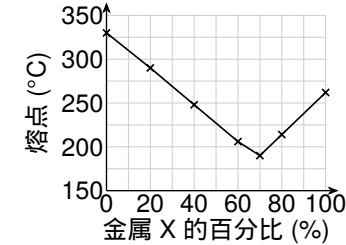
Melting points of alloys



Mixtures of two metals, X and Y.

- 1 What is the melting point of pure X? Of pure Y?
- 2 Which mixture has the lowest melting point?
- 3 Describe the trend from 0% to 70% X.

合金的熔点



两种金属 X 和 Y 的混合物。

- 1 纯 X 的熔点是多少？纯 Y 呢？
- 2 哪个混合比例的熔点最低？
- 3 描述从 0% 到 70% X 的变化趋势。

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1 pure X 262 °C; pure Y 330 °C

2 70 % X, about 190 °C

3 the melting point falls steadily as the percentage of X increases

Practice

Q1 Name the heat treatment that makes steel hard but brittle. **[A]**

Q2 Brass is an alloy of copper and zinc. Explain why brass is harder than pure copper. **[M]**

Q3 A car spring and a car door panel are both made of steel. Explain which heat treatment each one should have. **[E]**

Practice 练习

Q1 哪种热处理让钢变硬但变脆？ **[A]**

Q2 brass（黄铜）是铜和锌的合金。解释为什么黄铜比纯铜硬。 **[M]**

Q3 汽车的弹簧和车门板都用钢做。解释它们各应该做哪种热处理。 **[E]**

Q1 quenching

Q2 zinc atoms are a different size → disrupt the layers → layers cannot slide easily

Q3 spring: tempering (tough, springy); door panel: annealing (soft, easy to press into shape)

Quick review: Corrosion and rusting

True or false?

- 1 Rusting is a type of corrosion.
- 2 Any metal can rust.
- 3 Salt water makes iron rust faster.
- 4 Gold corrodes quickly in air.
- 5 Corrosion is a chemical reaction of a metal with substances such as oxygen and water.

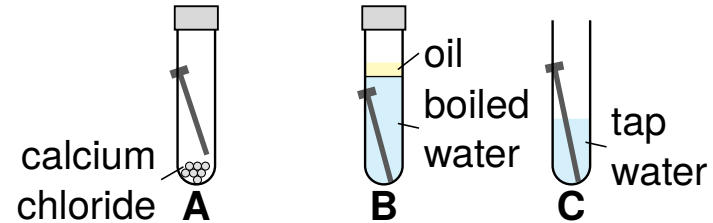
Quick review 快速回忆：腐蚀和生锈

判断对错 (true / false):

- 1 rusting (生锈) 是 corrosion (腐蚀) 的一种。
- 2 任何金属都会生锈。
- 3 盐水让铁生锈得更快。
- 4 金在空气里很快被腐蚀。
- 5 腐蚀是金属和氧气、水等物质发生的化学反应。

1 true 2 false (只有 iron/steel 会 rust) 3 true 4 false 5 true

Rusty nail experiment (1): the set-up



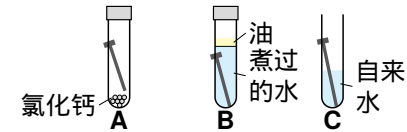
Iron nails, left for 3 days.

A The calcium chloride removes _____.

B Boiling removes _____ from the water. The oil stops _____.

C The nail has both _____ and _____.

生锈实验 (1): 装置



铁钉放 3 天。calcium chloride = 氯化钙。

A 氯化钙吸走了 _____。

B 煮沸赶走了水里的 _____；油阻止 _____。

C 钉子既有 _____ 又有 _____。

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A water (moisture)
B oxygen; air getting back in
C water and oxygen

Rusty nail experiment (2): write it in this order

Results after 3 days: A _____ B _____ C _____

- 1 Only tube _____ rusted.
- 2 Tube A had air but no _____.
- 3 Tube B had water but no _____.
- 4 So iron needs _____ to rust.

A: C rusted **M:** + what A and B lacked **E:** the conclusion from all three

生锈实验 (2): 按顺序写

3 天后的结果: A _____ B _____ C _____

- 1 只有 _____ 号试管生锈。
- 2 A 管有空气, 但没有 _____。
- 3 B 管有水, 但没有 _____。
- 4 所以铁生锈需要 _____。

A: C 生锈 **M:** + A 和 B 各缺什么 **E:** 用三支试管得出结论

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结果: A no rust B no rust C rust
1 C 2 water 3 oxygen 4 both water and oxygen

Spot the error (5)

Steel paper clips are sealed in three tubes: P with calcium chloride (dry air), Q in boiled water under a layer of oil, R in tap water with air above. Describe and explain the results.

Sam's answer

P: no rust, because there is no air. Q: a little rust, because it has water. R: lots of rust. So iron only needs water to rust.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error (5) 找错误

钢制回形针分别封在三支试管里：P 有氯化钙（干燥空气），Q 在煮过的水里、上面有一层油，R 在自来水里、上面有空气。描述并解释结果。

Sam's answer

P: no rust, because there is no air. Q: a little rust, because it has water. R: lots of rust. So iron only needs water to rust.

Sam 的答案（英文）有 3 个错误。找出每一个，改正，并说明为什么会丢分。

1 P has air; it does not rust because it has no water

2 Q does not rust: boiled water has no oxygen and the oil keeps air out

3 iron needs both water and oxygen

R 那一句是对的

Preventing rust

Object	Method	Why this method
steel fence	_____	large, cheap, easy to redo
door hinge	oil or grease	_____
rubbish bin	_____	zinc coating
kitchen sink	_____	never needs a coating

Every method keeps _____ and _____ away from the iron.

防止生锈

物体	方法	为什么用这种方法
钢栅栏	_____	面积大、便宜、容易重做
门铰链	上油或润滑脂	_____
垃圾桶	_____	镀一层锌
厨房水槽	_____	永远不需要涂层

每种方法都是让 _____ 和 _____ 接触不到铁。

钢栅栏：painting
门铰链：moving parts, paint would rub off
垃圾桶：galvanising
厨房水槽：stainless steel
最后一句：water and oxygen

Practice

Q1 Name the two substances iron needs in order to rust. **[A]**

Q2 Explain why a car in a seaside town rusts faster than the same car far inland. **[M]**

Q3 Plan an experiment to find out whether salt water makes iron nails rust faster than tap water. Say what you would change, what you would measure, and what you would keep the same. **[E]**

Practice 练习

Q1 铁生锈需要哪两种物质？ **[A]**

Q2 为什么海边小镇的汽车，比内陆同样的汽车生锈更快？ **[M]**

Q3 设计一个实验：盐水是否比自来水让铁钉生锈更快？说出你改变什么、测量什么、保持什么不变。 **[E]**

Q1 water and oxygen

Q2 sea air carries salt, and salt speeds up rusting

**Q3 change: tap water or salt water; measure: amount of rust (or mass) after 3 days
keep the same: nail type and size, volume of water, temperature, time; repeat**

Quick review: Word equation patterns

- 1 metal + oxygen $\rightarrow$ _____
- 2 metal + water $\rightarrow$ _____ + _____
- 3 metal + hydrochloric acid $\rightarrow$ _____ + _____
- 4 Test for hydrogen: a lit splint gives a _____.

Quick review 快速回忆：文字方程式的规律

- 1 金属 + oxygen (氧气) $\rightarrow$ _____
- 2 金属 + water (水) $\rightarrow$ _____ + _____
- 3 金属 + hydrochloric acid (盐酸) $\rightarrow$ _____ + _____
- 4 检验 hydrogen (氢气): 点燃的木条发出 _____。

- 1 metal oxide
- 2 metal hydroxide + hydrogen
- 3 metal chloride + hydrogen
- 4 squeaky pop

Writing word equations

(a) Iron wool burns in oxygen.

iron + oxygen → _____

(b) Lithium fizzes in cold water.

lithium + water → _____ + _____

(c) Aluminium powder in dilute hydrochloric acid.

(d) Gold in dilute hydrochloric acid.

写文字方程式

(a) 钢丝绒在氧气里燃烧。

iron + oxygen → _____

(b) lithium (锂) 在冷水里冒泡。

lithium + water → _____ + _____

(c) 铝粉放进稀盐酸。

(d) 金放进稀盐酸。

Metals p.12, p.18–19

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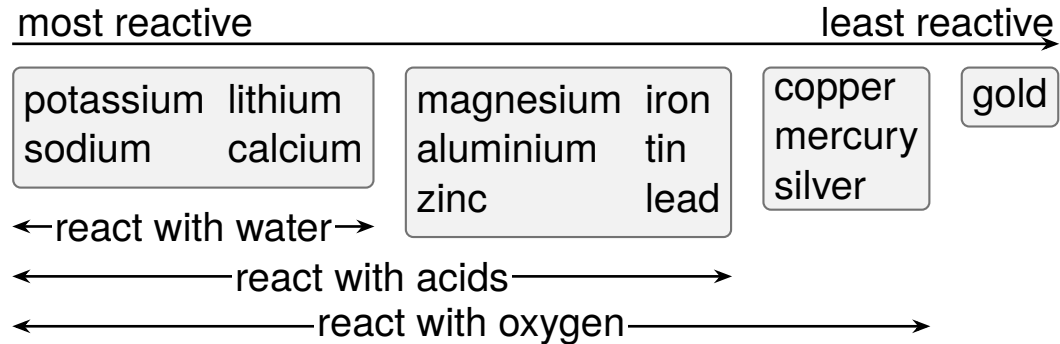
(a) iron oxide

(b) lithium hydroxide + hydrogen

(c) aluminium + hydrochloric acid → aluminium chloride + hydrogen

(d) no reaction

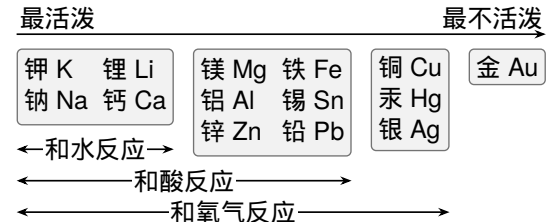
The reactivity series



1 Would tin react with dilute acid?

2 Why is gold found in the ground as pure metal, but iron is not?

The reactivity series 金属活动性顺序



1 tin (锡) 会和稀酸反应吗?

2 为什么地下能找到纯的金, 却找不到纯的铁?

1 yes (tin 在 react with acids 的范围里, 很慢)

2 gold is very unreactive, so it stays as the metal; iron is more reactive, so it is found as compounds (ores)

Spot the error (6)

Complete each word equation, or write “no reaction”.

Sam's answer

1. calcium + oxygen makes calcium oxygen
2. iron + hydrochloric acid makes iron chloride + water
3. potassium + water makes potassium hydroxide + hydrogen
4. silver + hydrochloric acid makes silver chloride + hydrogen

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error (6) 找错误

完成每个文字方程式，或者写 “no reaction”（不反应）。

Sam's answer

1. calcium + oxygen makes calcium oxygen
2. iron + hydrochloric acid makes iron chloride + water
3. potassium + water makes potassium hydroxide + hydrogen
4. silver + hydrochloric acid makes silver chloride + hydrogen

Sam 的答案（英文）有 3 个错误。找出每一个，改正，并说明为什么会丢分。

- 1 calcium oxide, 不是 calcium oxygen
- 2 iron chloride + hydrogen, 不是 water
- 3 silver + hydrochloric acid → no reaction
- 第 3 条（potassium）是对的

Practice

Q1 Write the word equation for aluminium reacting with oxygen. **[A]**

Q2 Four metals are added to dilute acid. Put them in order of reactivity, most reactive first, and give your evidence. **[M]**

P many bubbles, quickly

Q no bubbles

R a few bubbles, slowly

S reacts even with cold water

Q3 Describe an experiment with dilute hydrochloric acid to show that zinc is more reactive than copper. Give the result you expect. **[E]**

Practice 练习

Q1 写出铝和氧气反应的文字方程式。 **[A]**

Q2 四种金属放进稀酸。按活泼程度从高到低排列，并给出证据。 **[M]**

P 气泡很多、很快

Q 没有气泡

R 气泡很少、很慢

S 连冷水都能反应

Q3 描述一个用稀盐酸证明锌比铜活泼的实验，并说出你预期的结果。 **[E]**

Q1 aluminium + oxygen → aluminium oxide

Q2 S, P, R, Q —S reacts even with cold water; P many bubbles fast; R few, slow; Q none

Q3 same size zinc and copper, same volume and concentration of acid: zinc fizzes (hydrogen), copper no reaction

Ways to lose marks: Metals

L10 Density = mass $\div$ volume, unit g/mL or g/cm³.

L11 Volume = final reading – first reading.

L12 Property + why it matters + comparison.

L13 Different-sized atoms, layers cannot slide.

L14 Rusting needs water **and** oxygen.

L15 Oxide; hydroxide + hydrogen; chloride + hydrogen.

最常见的丢分点: Metals

L10 密度 = 质量 $\div$ 体积, 单位 g/mL 或 g/cm³。

L11 体积 = 放入后的读数 – 放入前的读数。

L12 性质 + 为什么重要 + 对比。

L13 原子大小不同, 层不能滑动。

L14 生锈需要水 **和** 氧气。

L15 oxide; hydroxide + hydrogen; chloride + hydrogen。

Metals p.1

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让她每条举例: **L10、L11 Sam 的螺栓; L13 黄铜; L14 回形针; L15 Sam 的方程式**

Summary: Metallic materials

Properties decide uses: say the property, why it matters, and compare.

Density = mass $\div$ volume; displacement gives the volume.

Alloys are harder: different-sized atoms stop the layers sliding.

Rust needs water and oxygen; reactivity decides how metals react.

Summary 总结: Metallic materials

性质决定用途：说出性质、为什么重要，再比较。

密度 = 质量 $\div$ 体积；用排水法测体积。

合金更硬：大小不同的原子让层不能滑动。

生锈需要水和氧气；活泼程度决定金属怎样反应。

本页没有题目。先闭眼说一遍，再看屏幕。

Classwork: Metallic materials

On your classwork sheet, about 10 minutes:

- D** Properties and density
- E** Alloys and heat treatment
- F** Corrosion and reactions, with an experiment to plan

Show every step of a calculation, with units.

Classwork 课堂练习: Metallic materials

在课堂练习纸上做，大约 10 分钟：

- D** 性质和密度
 - E** 合金和热处理
 - F** 腐蚀和反应，有一个实验要设计
- 计算题每一步都写出来，带单位。

Classwork Part D–F

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做 Classwork Part D–F，约 10 分钟；答案在 [classwork_answers.pdf](#)
Part D 密度题：2.7 g/cm³ (aluminium)

Mini mock

17 minutes · both units · on the paper

Answer every part.

Each question is marked A, M or E.

Explain questions: write it in the order you practised today.

Mini mock 小模考

17 分钟 · 两个单元 · 写在卷子上

每一问都要回答。

每题按 A、M、E 评分。

Explain 题：按今天练习的顺序写。

发 **mock.pdf**，计时 17 分钟，不提示；答案在 **mock_answers.pdf**

Homework

hw1 Green Machines, about 40 minutes

hw2 Metallic materials, about 40 minutes

Also: Metals booklet p.10–11, draw the solder graph and answer Q1–2.

Do them on two different days before the exam.

Homework 作业

hw1 Green Machines, 大约 40 分钟

hw2 Metallic materials, 大约 40 分钟

另外：Metals booklet 第 10–11 页，画焊锡的图，并回答 Q1–2。
在考试前，分两天做完。

hw1、hw2 各约 40 分钟；答案在 [hw1_answers.pdf](#)、[hw2_answers.pdf](#)